

Supplementary Material for “Grouped Proportional Hazards under Coarse Time: Information Loss and Baseline Estimation”

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S1 Proof of the profiled small-bin expansion

Let

$$F(a, b, c) = c - \frac{b^2}{a}. \quad (\text{S1})$$

For interval t , write

$$M_j^{(t)} = \mathbb{E}\{v(X)^j W_t(X)\}, \quad (\text{S2})$$

and

$$N_j^{(t)} = \mathbb{E}\left[v(X)^j S(a_{t-1} \mid X)\{1 - e^{-u_t(X)}\}\right] = \int_{I_t} m_j(s) ds. \quad (\text{S3})$$

Write the continuous and grouped profiled contributions over interval I_t as

$$\mathcal{I}_{c,t} = \int_{I_t} F\{m_0(s), m_1(s), m_2(s)\} ds, \quad \mathcal{I}_{g,t} = F(M_0^{(t)}, M_1^{(t)}, M_2^{(t)}). \quad (\text{S4})$$

Then

$$\begin{aligned} \mathcal{I}_{\infty}^* - \mathcal{I}_T^* &= \sum_t \left[\int_{I_t} F\{m(s)\} ds - F\{N^{(t)}\} \right] \\ &\quad + \sum_t \left[F\{N^{(t)}\} - F\{M^{(t)}\} \right]. \end{aligned} \quad (\text{S5})$$

For the second sum, Eq. (22) implies

$$N_j^{(t)} - M_j^{(t)} = \mathbb{E}\left[v(X)^j S(a_{t-1} \mid X) \psi\{u_t(X)\}\right]. \quad (\text{S6})$$

The first-order perturbation of F is

$$\delta F = \delta N_2 - 2\bar{v} \delta N_1 + \bar{v}^2 \delta N_0. \quad (\text{S7})$$

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Using Eq. (27), the second sum in Eq. (S5) is therefore

$$\frac{\Delta^2}{12} \int_0^\tau \mathbb{E} \{ [v(X) - \bar{v}(s)]^2 S(s | X) \lambda(s | X)^3 \} ds + O(\Delta^3). \quad (\text{S8})$$

For the first sum, let

$$\bar{m}_j = \Delta^{-1} \int_{I_t} m_j(s) ds, \quad \tilde{m}(s) = m(s) - \bar{m}. \quad (\text{S9})$$

The function F is homogeneous of degree one, so

$$F\{N^{(t)}\} = \Delta F(\bar{m}). \quad (\text{S10})$$

A second-order Taylor expansion around $\bar{m}$ gives

$$\int_{I_t} F\{m(s)\} ds - \Delta F(\bar{m}) = \frac{1}{2} \int_{I_t} \tilde{m}(s)^\top \nabla^2 F(\bar{m}) \tilde{m}(s) ds + O(\Delta^4). \quad (\text{S11})$$

For a perturbation $(\tilde{a}, \tilde{b}, \tilde{c})$, the Hessian quadratic form is

$$(\tilde{a}, \tilde{b}, \tilde{c}) \nabla^2 F(a, b, c) (\tilde{a}, \tilde{b}, \tilde{c})^\top = -\frac{2}{a} \left(\tilde{b} - \frac{b}{a} \tilde{a} \right)^2. \quad (\text{S12})$$

Since $\bar{v} = m_1/m_0$,

$$m'_1 - \bar{v} m'_0 = m_0 \bar{v}'. \quad (\text{S13})$$

The leading within-bin linear term in Eq. (S12) therefore gives

$$\int_{I_t} F\{m(s)\} ds - F\{N^{(t)}\} = -\frac{\Delta^3}{12} m_0(a_{t-1}) \{\bar{v}'(a_{t-1})\}^2 + O(\Delta^4). \quad (\text{S14})$$

Summing over $T = \tau/\Delta$ intervals gives

$$-\frac{\Delta^2}{12} \int_0^\tau m_0(s) \{\bar{v}'(s)\}^2 ds + O(\Delta^3). \quad (\text{S15})$$

Combining Eqs. (S8) and (S15) proves Eq. (37).

S2 All 34 main baseline-decomposition configurations

Tables S1 and S2 report the complete main sweep. The modal event masses were computed from failures only. The standard errors are corrected repeated-split standard errors from 20 shared 70/30 splits.

Table S1 R1 configurations. Positive D values favour grouped joint maximum likelihood.

Configuration	$M_T^{(E)}$	D_B	SE_B	D_{KP}	SE_{KP}
metabric, $T = 4$	0.2502	0.00761	0.00485	-0.00024	0.00022
metabric, $T = 6$	0.2004	0.00394	0.00333	-0.00013	0.00012
metabric, $T = 10$	0.1006	0.00164	0.00232	-0.00008	0.00007
metabric, $T = 20$	0.0517	-0.00002	0.00135	-0.00004	0.00004
nwtco, $T = 4$	0.2539	-0.00005	0.00041	-0.00002	0.00002
nwtco, $T = 6$	0.2102	-0.00009	0.00029	-0.00001	0.00001
nwtco, $T = 10$	0.1086	-0.00008	0.00017	-0.00001	0.00001
nwtco, $T = 20$	0.0595	-0.00009	0.00008	-0.00000	0.00001
fchain, $T = 4$	0.2503	0.00151	0.00253	-0.00194	0.00238
fchain, $T = 6$	0.2006	0.00044	0.00229	-0.00141	0.00203
fchain, $T = 10$	0.1010	0.00036	0.00062	-0.00025	0.00030
fchain, $T = 20$	0.0507	0.00002	0.00030	-0.00010	0.00012
support-pycox, $T = 4$	0.2614	0.00892	0.00173	-0.00006	0.00007
support-pycox, $T = 6$	0.2071	0.00416	0.00116	-0.00003	0.00003
support-pycox, $T = 10$	0.1175	0.00170	0.00077	-0.00001	0.00001
support-pycox, $T = 20$	0.0611	0.00042	0.00045	-0.00000	0.00001

Table S2 R2 configurations. Positive D values favour grouped joint maximum likelihood.

Configuration	$M_T^{(E)}$	D_B	SE_B	D_{KP}	SE_{KP}
drsa/clinic, $T = 5$	0.8374	0.10464	0.00762	-0.00069	0.00060
drsa/clinic, $T = 10$	0.7095	0.05785	0.00643	-0.00011	0.00104
drsa/clinic, $T = 25$	0.5406	0.02269	0.00422	-0.00022	0.00023
drsa/clinic, $T = 50$	0.4040	0.00842	0.00252	-0.00010	0.00012
support2/slos, $T = 6$	0.4912	0.03928	0.00446	-0.00027	0.00045
support2/slos, $T = 10$	0.3306	0.01839	0.00305	-0.00018	0.00024
support2/slos, $T = 15$	0.2818	0.01019	0.00233	-0.00013	0.00013
support2/slos, $T = 20$	0.2179	0.00624	0.00177	-0.00008	0.00008
support2/slos, $T = 30$	0.1527	0.00257	0.00130	-0.00004	0.00004
support2/slos, $T = 60$	0.0776	0.00041	0.00069	-0.00001	0.00001
sparcs/drg302, $T = 6$	0.9032	0.29336	0.00261	0.00135	0.00128
sparcs/drg302, $T = 10$	0.6679	0.14313	0.00264	-0.00031	0.00071
sparcs/drg302, $T = 15$	0.7191	0.16779	0.00350	0.00045	0.00025
sparcs/drg302, $T = 30$	0.5454	0.06707	0.00232	0.00024	0.00060
drsa/music, $T = 6$	0.4670	0.02025	0.00122	-0.00000	0.00001
drsa/music, $T = 12$	0.2417	0.00736	0.00082	-0.00000	0.00000
drsa/music, $T = 30$	0.1020	0.00134	0.00040	-0.00000	0.00000
drsa/music, $T = 60$	0.0534	0.00028	0.00022	-0.00000	0.00000

S3 Secondary flexibility analysis

The original study also asked whether a learned Kolmogorov–Arnold representation improved prediction beyond ordinary smooth additive effects. Because this question is separate from the baseline-estimation result, it is reported as a secondary analysis. Kolmogorov–Arnold networks parameterize univariate edge functions, often with splines (Liu et al. 2025); recent survival applications include CoxKAN, KAN-AFT, SurvKAN and KAPLAN (Knottenbelt et al. 2025; Jose et al. 2025; Mastroleo et al. 2026; Boulitsakis Logothetis et al. 2026).

Five cohorts were analysed with 20 seeds. For each continuous covariate, a cubic B-spline additive model was considered over $q \in \{4, 6, 8, 10, 12\}$ basis

dimensions. The value of q was selected using a validation subset carved from the training data, not from the test data (de Boor 1978; Hastie and Tibshirani 1990). Comparators included a penalized P-spline, a KAN matched to the selected additive function space, an independently tuned KAN, a depth-2 KAN, and an additive model augmented by selected pairwise interactions. All models used the same grouped survival likelihood.

Evaluation included held-out negative log-likelihood, Antolini’s time-dependent concordance (Antolini et al. 2005), Uno’s censoring-robust concordance (Uno et al. 2011), integrated Brier score (Graf et al. 1999), and the integrated calibration index for survival predictions (Austin et al. 2020). Tables S3 and S4 report paired performance differences for the matched KAN relative to the validation-selected additive spline model. Every difference is oriented so that a positive value favours the matched KAN.

Table S3 Matched KAN relative to the validation-selected additive spline model over 20 seeds: likelihood and discrimination. Entries are paired performance differences $\pm$ their reported uncertainty, oriented so that positive values favour the KAN. None was resolved.

Cohort	NLL	Antolini C	Uno C
rotgbsg	0.00489 ± 0.00798	0.00100 ± 0.00528	0.00121 ± 0.00520
metabric	0.00686 ± 0.00482	0.00265 ± 0.00311	0.00196 ± 0.00242
nwtco	0.00216 ± 0.00264	0.00043 ± 0.00396	0.00054 ± 0.00410
flchain	0.01302 ± 0.03177	0.00002 ± 0.00120	0.00021 ± 0.00117
support-pycox	0.00096 ± 0.00275	-0.00165 ± 0.00231	-0.00154 ± 0.00219

Table S4 Matched KAN relative to the validation-selected additive spline model over 20 seeds: integrated Brier score and integrated calibration index. Entries are paired performance differences $\pm$ their reported uncertainty, oriented so that positive values favour the KAN. None was resolved.

Cohort	IBS	ICI
rotgbsg	-0.00018 ± 0.00168	-0.00034 ± 0.00783
metabric	0.00096 ± 0.00080	0.00340 ± 0.00745
nwtco	-0.00053 ± 0.00046	0.00090 ± 0.00264
flchain	-0.00023 ± 0.00029	0.00022 ± 0.00212
support-pycox	-0.00024 ± 0.00042	0.00100 ± 0.00436

The matched comparisons do not identify an architecture-specific KAN advantage. The support-pycox interaction model did produce a small resolved improvement over the additive spline model in Antolini concordance (0.00274 ± 0.00140) and integrated Brier score (0.00066 ± 0.00024). The secondary analysis therefore supports a narrower conclusion: nonlinear main effects can be represented adequately by conventional smooth terms in these cohorts, while non-additive structure can matter in at least one dataset. It does not establish a general superiority of one architecture.

Reference note

References cited in this supplement are listed in the main article.